

HUYEN TRAN ALEXANDRA BANK, Viet Nam

WMO: 489190

Lat: 8.017N

Lon: 110.617E

Elev: 19

StdP: 101.10

Time Zone: 7.00 (E07)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	24.3	24.8	19.8	14.5	25.7	20.3	15.0	25.9	N/A	N/A	N/A	N/A	N/A	N/A	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	1.9	31.2	26.5	30.7	26.4	30.3	26.3	27.3	29.8	27.1	29.7	26.8	29.4	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.6	22.2	29.2	26.2	21.7	29.0	26.0	21.4	28.9	86.6	30.0	85.4	29.9	84.4	29.6	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	22.7	33.6	1.1	2.7	22.0	35.5	21.4	37.1	20.8	38.6	20.0	40.5
			WB	21.0	28.0	0.7	0.7	20.5	28.5	20.1	28.8	19.8	29.2	19.3	29.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	26.5	26.3	27.3	28.6	29.1	28.6	28.1	28.1	28.0	28.0	27.5	27.0
	DBStd	1.24	0.93	0.67	0.84	1.06	1.06	1.01	0.90	0.80	0.96	0.87	0.86	1.06
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6483	511	457	535	557	593	558	560	560	541	558	527	527
	CDD18.3	3442	252	224	277	307	335	308	302	302	291	299	277	269
	CDH23.3	39162	2360	2023	2913	3738	4331	3849	3601	3559	3427	3474	3111	2774
CDH26.7	11296	238	108	531	1374	1863	1477	1172	1122	1082	1036	795	498	
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	2069	96	52	53	42	135	205	250	184	230	209	272	318
	PrecMax	2534	388	163	323	107	307	416	419	287	516	487	501	854
	PrecMin	1489	2	0	0	0	21	53	77	99	64	58	95	62
	PrecStd	283	88	54	80	30	73	85	80	59	110	102	110	205
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.1	28.2	30.2	31.6	32.2	31.7	30.6	30.2	31.1	31.1	30.0	29.7
		MCWB	25.2	25.0	26.1	26.6	26.8	26.5	26.2	26.2	26.2	26.2	26.0	25.7
	2%	DB	28.6	27.7	29.2	30.7	31.2	30.7	30.0	29.6	30.2	30.1	29.4	29.2
		MCWB	25.1	24.8	25.6	26.2	26.6	26.5	26.1	26.0	25.9	25.9	25.8	25.6
	5%	DB	28.2	27.5	28.6	30.2	30.7	30.2	29.6	29.2	29.6	29.5	29.1	28.9
		MCWB	25.1	24.7	25.3	26.1	26.5	26.4	26.0	25.9	25.8	25.7	25.8	25.5
	10%	DB	27.9	27.2	28.2	29.7	30.3	30.0	29.2	29.1	29.2	29.2	28.9	28.5
		MCWB	25.0	24.6	25.1	25.9	26.4	26.4	25.9	25.9	25.7	25.6	25.7	25.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.2	26.6	27.0	27.5	27.7	27.7	27.0	27.2	26.7	26.7	26.7	26.7
		MCDB	27.7	27.4	28.7	29.7	30.9	30.0	29.3	29.2	29.5	29.4	28.6	28.4
	2%	WB	25.8	25.9	26.3	27.0	27.2	27.4	26.7	26.7	26.5	26.4	26.4	26.3
		MCDB	27.5	27.1	28.3	29.6	30.3	29.8	29.1	28.9	29.2	29.1	28.5	28.2
	5%	WB	25.6	25.5	25.9	26.7	27.1	27.0	26.5	26.5	26.2	26.2	26.2	26.1
		MCDB	27.3	27.0	28.1	29.4	30.1	29.6	29.0	28.8	28.9	28.9	28.4	28.1
	10%	WB	25.3	25.2	25.6	26.4	26.8	26.7	26.2	26.2	26.1	26.0	26.0	25.8
		MCDB	27.2	26.9	27.8	29.2	29.8	29.4	28.9	28.6	28.8	28.7	28.3	27.9
Mean Daily Temperature Range	5% DB	MDBR	1.1	1.1	1.3	1.7	1.9	1.9	1.9	1.6	2.1	2.0	1.9	1.3
		MCDBR	1.3	1.3	1.9	2.3	2.3	2.1	2.1	1.7	2.4	2.5	2.2	1.5
		MCWBR	0.7	0.8	1.1	1.2	1.1	1.1	1.1	1.0	1.1	1.1	1.0	0.8
	5% WB	MCDBR	1.3	1.1	1.6	2.0	2.0	1.8	1.9	1.5	2.2	2.2	1.9	1.5
		MCWBR	0.8	0.8	1.0	1.2	1.2	1.3	1.4	1.1	1.2	1.2	1.1	1.0
Clear-Sky Solar Irradiance	taub	0.475	0.443	0.439	0.422	0.414	0.428	0.440	0.455	0.441	0.445	0.460	0.518	
	taud	2.295	2.396	2.434	2.487	2.531	2.506	2.457	2.396	2.427	2.427	2.368	2.143	
	Ebn at Noon	841	882	886	888	877	854	847	847	870	869	849	791	
	Edh at Noon	134	124	120	113	105	106	112	122	119	118	123	153	
All-Sky Solar Radiation	RadAvg	4.44	5.69	6.15	6.48	5.82	5.03	4.88	5.17	4.99	4.96	4.39	3.65	
	RadStd	0.66	0.60	0.71	0.43	0.54	0.33	0.49	0.39	0.57	0.56	0.34	0.75	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page